

Go Language Fundamentals

AI Tutor - Golang - Lesson 1

Introduction

Go (or Golang) is an open-source programming language created at Google, designed for simplicity, fast compilation, and strong built-in support for concurrent programs. It's widely...

Explanation

Go (or Golang) is an open-source programming language created at Google, designed for simplicity, fast compilation, and strong built-in support for concurrent programs. It's widely used for backend servers, APIs, and cloud infrastructure. Every Go program starts with a 'package main' declaration and a 'main' function, the entry point where execution begins. Go compiles directly to a single, fast, native binary, which makes deployment simple.

Key Points

- Go programs start with 'package main' and a 'main' function.
- Go compiles to a single, fast, native binary.
- Go has built-in support for concurrency via goroutines.
- Go's simple, readable syntax is strongly typed.

Examples

- `package main; import "fmt"; func main() { fmt.Println("Hello, Go!") }`
- The Gin framework is commonly used to build REST APIs in Go.

Summary

Go is a simple, fast-compiling language built for concurrent backend systems, where every program starts from a 'main' function and compiles to a single native binary.

Practice Questions

- 1 What is the name of the function where every Go program begins execution?
- 2 Name one reason developers choose Go for backend systems.
- 3 What does Go compile a program into?

Quiz

Q1. Which function is the entry point of every Go program?

a) start() | b) main() | c) init() | d) run()

Q2. What does Go compile a program into?

a) A virtual machine bytecode | b) A single native binary | c) An interpreted script | d) A Java archive

Q3. What Go feature provides built-in support for concurrent programs?

a) Threads | b) Goroutines | c) Callbacks | d) Promises